

Cross-platforming “School life metaverse” user experience

Takuma Kato
Digital Hollywood University
Chiyoda, Tokyo, Japan
takuma.katou42@gmail.com

Miku Takemasa
Wakayama University
Wakayama, Wakayama, Japan
takemasa.wadai@gmail.com

Tomosuke Nakano
Meiji University
Nakano, Tokyo, Japan
tomosuke.nakano@gmail.com

Yusuke Yamazaki
Tokyo Institute of Technology
Yokohama, Kanagawa, Japan
yus988@hapbeat.com

Takanori Horibe
Meiji University
Nakano, Tokyo, Japan
takanori.horibe@gmail.com

Akihiko Shirai
GREE VR Studio Laboratory
Minato, Tokyo, Japan
shirai@mail.com



Figure 1: Screenshots from the “Metaver-school” project. Guitar playing by AI Fusion (left), bunch of avatar benchmark on PCVR (mid), and multi-player cross-platform band play with bunch of virtual gifting (Right).

CCS CONCEPTS

• Computing methodologies → Procedural animation; • Human-centered computing → Empirical studies in interaction design.

KEYWORDS

metaverse, user experience, avatar, real-time animation, music

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1 “METAVER-SCHOOL”, AN ATTRACTIVE SCHOOL LIFE VRSNS FOR TEENAGERS

The development of new user experiences (UX) in a metaverse, with a communication space and 3D avatars can contribute to extending human beings in the context of computer graphics, interactive techniques and their applications. This study, “Metaver-School”, focuses on the potential value and comfortable communication of UX in a near-future VRSNS UX exploration project for teenagers across smartphones, current standalone HMDs and full-spec PCVR,

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with a variety of avatars, which can be experienced interchangeably across different platforms [Laboratory 2022a].

2 BENCHMARKS USING DIVERSITY AVATARS

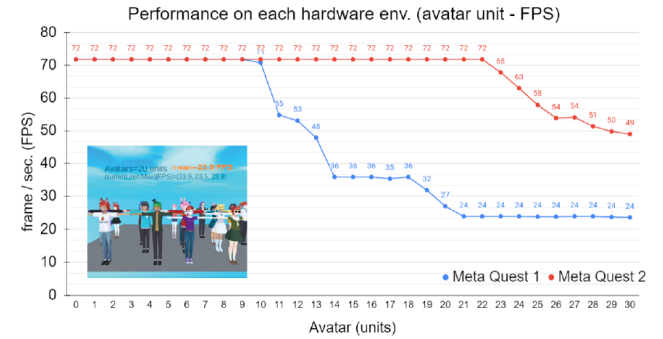


Figure 2: Benchmarking using various avatars using BRP Unlit shader on Quest1 and Quest2

In this context, the project focused on having benchmarks of current smartphone metaverse service “REALITY” compatible avatars¹ [Shirai 2019] on current standalone HMDs like Meta Quest1/2². We first investigated how many avatars could be rendered on current Head Mounted Displays (HMDs). Representing school scenes requires rendering many avatars simultaneously at a certain quality

¹“REALITY” is personal 3D avatar broadcasting communication service works on a smartphone. Released in 2018 and available in 63 countries (<https://reality.app>).

²In this paper, we use “Quest1” to refer to the first generation Oculus Quest.

with diversity that is currently served in REALITY. Japanese anime-style avatar expression commonly uses VRM's MToon and UnLit (not affected by lighting) shaders, and it runs in Unity's Built-in-Render Pipeline (BRP). We compared the Draw call in Quest with a modern Universal Render Pipeline (URP). VRM-MToon in BRP was 122, VRM-Unlit (BRP) was 66 and URP-UnLit was 38, with URP-UnLit having a lower load. On the other hand, when we compared frame rates in our testing environment by increasing avatars to 5, 10, 20 and 40, BRP and URP showed that 5 avatars are comfortable (FPS>60), although URP's SRP Batchier had an effect, and after that, there was no significant effect (FPS<40). This is an important finding for implementing a metaverse with diverse avatar representations on a variety of platforms. Traditional optimization methods in game engines may not apply to avatars with diverse clothing assets as an expression of the user's personality, such as in REALITY.

As a benchmark for cross-platform measurement, we fixed conditions as below: Unity ver 2020.3.26f1, Oculus Integration SDK ver. 35.0, REALITY avatar (Unity Humanoid, 30–32k polygons, BRP Unlit, 8–16 materials, animated with springbones) were loaded from 0 to 30 avatars in a one directional light scene. The comparison was made by switching the display state while on-memory. Less than 40 FPS was assumed to be unmeasurable. This allowed the number of avatars that can be rendered at more than 60 FPS when rendering REALITY avatars in Quest 1 and Quest 2 to be verified. In Quest 2, up to 23 avatars could be rendered at more than 60 FPS, however in Quest 1, the maximum number of avatars that could achieve 60 FPS was ten avatars or less.

3 AI FUSION: COLLABORATIVE BAND PLAY IN CROSS-PLATFORM METAVERSE

"AI Fusion" is a procedural method to generate real-time music play animations for Humanoids using Standard MIDI and partial motion capture data like faces or hands [Laboratory 2022b]. High-quality real-time motion capture and sharing like VRChat has a chance to extend the UX but multi-user/multi-platform synchronization has more complicated issues. In this method, we solved these by merging procedural animations and the user's live action. We applied the technique for playable guitars, bass, drums, and keyboards, to generate generating finger animations from sheet music (MIDI file) and animation triggers. Fingering techniques were developed by experienced music students and under the guidance and supervision of professional performers. Multi-user interactive communication can be achieved even with stand-alone Quest devices by voice chat using Photon Voice, while transmitting the motion data (only three positions in 6DoF) that the user is acting out while communicating with the other clients (PC/Quest1/Quest2) in real-time.

4 CONCLUSION: VISUALIZATION PIPELINES FOR UXDEV-DRIVEN METAVERSE

We confirmed that experiences such as the music club in the school are important as experiences that cannot be done alone for the metaverse to be used on a daily basis. It is possible to accomplish virtual avatar karaoke by singing while controlling music played by an avatar but it can also be used as an avatar conference tool. It is more efficient having compatible UX between conventional

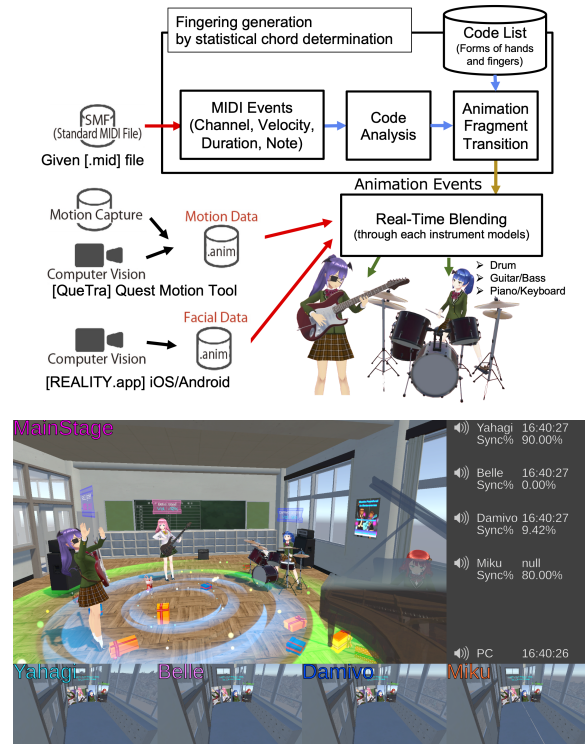


Figure 3: A proof of concept (PoC) of cross-platforming school life metaverse using AR Fusion and its music play. A maximum of four Quest clients can connect to each other. This screenshot was taken by a PC observing tool remotely.

smartphone UX and HMDs. Emotional expressions can be enhanced by recognition of eye gazing and voice analysis instead of linguistic expressions using button inputs.

We also used the Metaver-school project to create a tool "QueTra", which can save motions by Quest and converts them into .anim files on Unity. This tool also enabled us to build an animation pipeline. A short film series featuring the virtual band, "Meta Dreamers", was produced by a team of six people (mainly intern students) for one year as teasers and tutorial videos to make the real-time experience more engaging. A total of five short films were produced by Unity, which included a music video "Tsuyu-Hanabi" (<https://youtu.be/bU8QULxOTFU>). The proof of concept (PoC) results were recorded and published as short films. Individual techniques will inspire future users and contributors and be developed more deeply.

REFERENCES

- REALITY / GREE VR Studio Laboratory. 2022a. "Back to Metaver-School" - a blooming future of Metaverse in REALITY. YouTube Video. Retrieved April 24, 2022 from <https://youtu.be/1JID6XF2HNI>
- REALITY / GREE VR Studio Laboratory. 2022b. "Meta Dreamers" - UXDev by GREE VR Studio Laboratory - Tsuyu-Hanabi ft.nana. YouTube Video. Retrieved October 18, 2022 from <https://youtu.be/bU8QULxOTFU>
- Akihiko Shirai. 2019. REALITY: Broadcast Your Virtual Beings from Everywhere. In *ACM SIGGRAPH 2019 Appy Hour* (Los Angeles, California) (SIGGRAPH '19). Association for Computing Machinery, New York, NY, USA, Article 5, 2 pages. <https://doi.org/10.1145/3305365.3329727>